strata / strata_core architecture: two backends over one shared persistence + periodicity engine

strata_core — pure C++17 + Eigen (no ROS, no PCL) — gtest-tested

LayeredMap (per-cell evidence, int64 CellId)

log-odds occupancy: observeHit / observeMiss (l_hit/l_miss, clamped)
survival decay → toward "unknown" in each touched window
Schmitt trigger: graduate_prob / demote_prob thresholds
→ CellClass: Unknown | Transient | Periodic | Static
PeriodicityModel: FreMEn-lite Fourier fit (period_windows, n_harmonics)
amplitude ≥ a_min reclassifies Transient → Periodic

integrate()/tick()

integrate()/tick()

Grid2DBackend

GridMeta: W,H,res,origin_x,origin_y (fixed-size array)
hit → worldToGrid(x,y) [z dropped implicitly]
clear → Bresenham line (exact, on-grid ints)
toOccupancyGrid(): -1/0/50/75/100

Voxel3DBackend

int64 spatial hash id=(vx<<42)|(vy<<21)|vz, 21 bits per axis
hit → voxelId(x,y,z) [full 3D, no bounds check]
clear → ray-sample march (0.5-voxel step points)
staticPoints(): voxel centers → point cloud

MapBackend interface: integrate(Observation hits[MAP frame], origin) · tick() · closeWindows(k) · staticCellCount() · transientCellCount()

strata — ROS 2 node package (rclcpp / tf2 / PCL live only here)

TF buffer (external)

map ← sensor_frame
localizer/odom (e.g. prism_loc, not strata)

TF lookup(global_frame←sensor frame, 0.1s tol)

MappingNode (strata_node_main)

param "backend": grid2d | voxel3d
selects ONE backend at construction time
(no runtime hot-swap)
mtx_ serializes on{Scan,Points} vs onPublish/onSave

scan_adapter

/scan LaserScan
SensorDataQoS
→ Observation (6-DoF TF)
grid2d only

cloud_adapter

/points PointCloud2
SensorDataQoS
→ Observation (6-DoF TF)
voxel3d only

~/map

OccupancyGrid
transient_local+reliable
grid2d only

~/save_map (Trigger)

grid2d → .pgm + .yaml
voxel3d → .pcd

~/map_points

PointCloud2, QoS(1)
(volatile)
voxel3d only

strata never publishes map→odom and never subscribes /initialpose